

Revised
Merchant Transmission Interconnection
System Impact Study Report

For

PJM Generation Interconnection Request Queue
Position AB2-019

Erie West 345kV

February 2020

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner. In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed. The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

Revision from System Impact Study January 2020

The AB2-019 System Impact Study has been revised to reflect the correct Firm Transmission Injection Rights for the Y3-092 project.

General

ITC Lake Erie Connector, LLC, the Interconnection Customer (IC), has proposed a 28 MW uprate to their Queue Y3-092 Merchant Transmission line to connect Ontario IESO's transmission system to Mid-Atlantic Interstate Transmission's (or "MAIT") transmission system, in Erie County, PA. The installed facilities will have a capability of 1028 MW- with 1028 MW of this output being recognized by PJM as non-firm Transmission Withdrawal Rights (TWR). The installed facilities will also have a capability of 1000 MW Firm Transmission Injection Rights (TIR). (See chart below). The proposed Commercial Operation Date (COD) for this project is **March 31, 2024**.

Queue Position	Firm TWRs (MW)	Non-Firm TWRs (MW)	Firm TIRs (MW)	Non-Firm TIRs (MW)
Y3-092	0	1000	1000	0
AB2-019	0	28	0	0
Total	0	1028	1000	0

*See **Attachment 1** for the Project Location.*
*See **Attachment 2** for the One Line Diagram.*

Point of Interconnection (POI)

AB2-019 will share the same POI as the Y3-092 project which is an interconnection with the MAIT transmission system at the Erie West 345kV substation.

Cost Summary

The AB2-019 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrades	\$0
Non-Direct Connection Network Upgrades	\$0
Allocation for New System Upgrades	\$0
Contribution for Previously Identified Upgrades	\$0
Total Costs	\$0

Transmission Owner Scope of Work

Attachment Facilities

None

Direct Connection Facilities

None

Non-Direct Connection Facilities

None

Interconnection Customer Requirements

Uprate Queue AB2-019 is to meet the same interconnection customer requirements identified in the Y3-092 System Impact Study report dated April 2018.

Revenue Metering and SCADA Requirements

Uprate Queue AB2-019 is to meet the same interconnection customer requirements identified in the Y3-092 System Impact Study report dated April 2018.

Network Impacts

The Queue Project AB2-019 was evaluated as a 28 MW (Capacity 0.0 MW) withdrawal at Erie West 345kV substation in the Penelec area. Project AB2-019 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AB2-019 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Short Circuit

(Summary of impacted circuit breakers)

Not required for non-firm transmission withdrawal request.

Affected System Analysis & Mitigation

NYISO Impacts:

Please see the Y3-092 NYISO report. No additional issues with AB2-019.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

1. (PENELEC - PENELEC) The W3-099 TAP-26FOURMILE 230 kV line (from bus 903645 to bus 200928 ckt 1) loads from 99.91% to 100.39% (AC power flow) of its normal rating (488 MVA) for non-contingency condition. This project contributes approximately 2.32 MW to the thermal violation.

Light Load Analysis - 2020

Light Load not studied for non-firm transmission withdrawal requests.

System Reinforcements

Short Circuit

(Summary form of Cost allocation for breakers will be inserted here if any)

None

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

No mitigations required *(See Attachment 3 for an Executive Summary of the stability analysis results).*

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

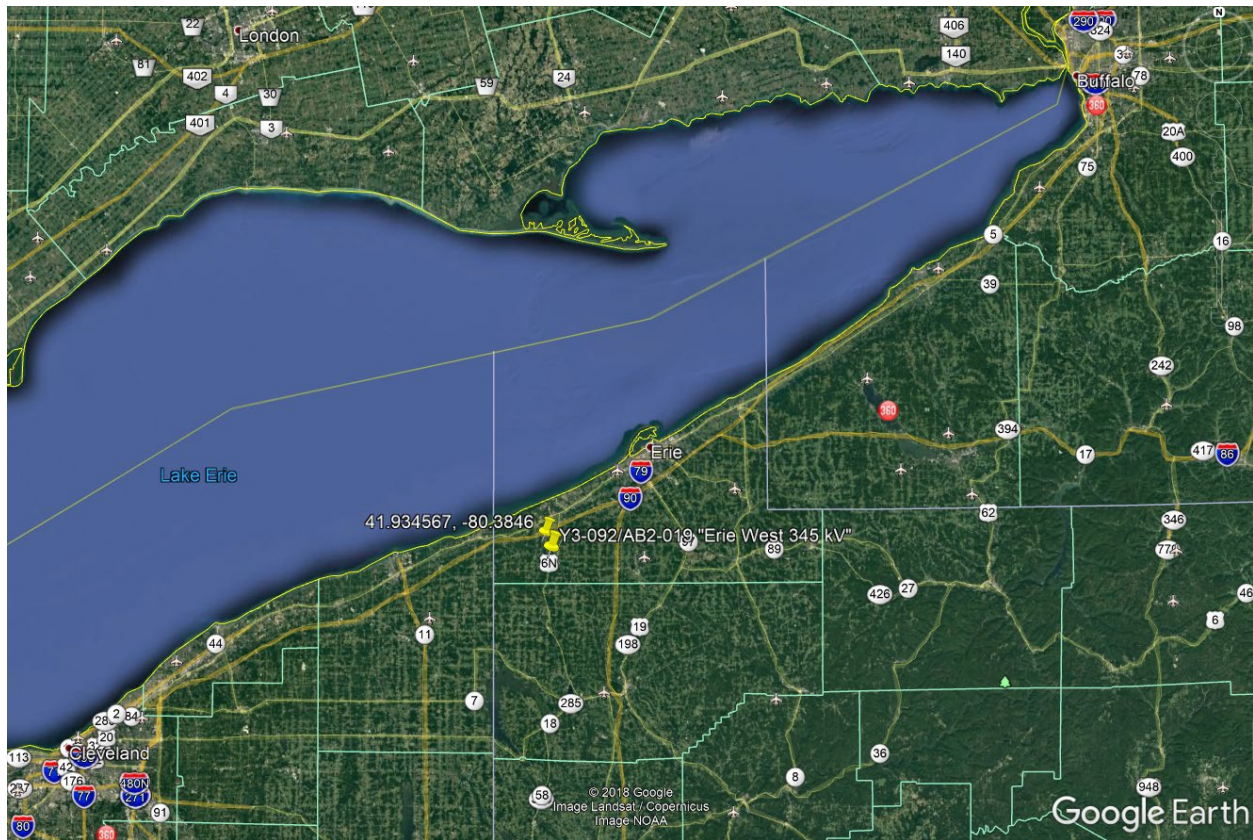
Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

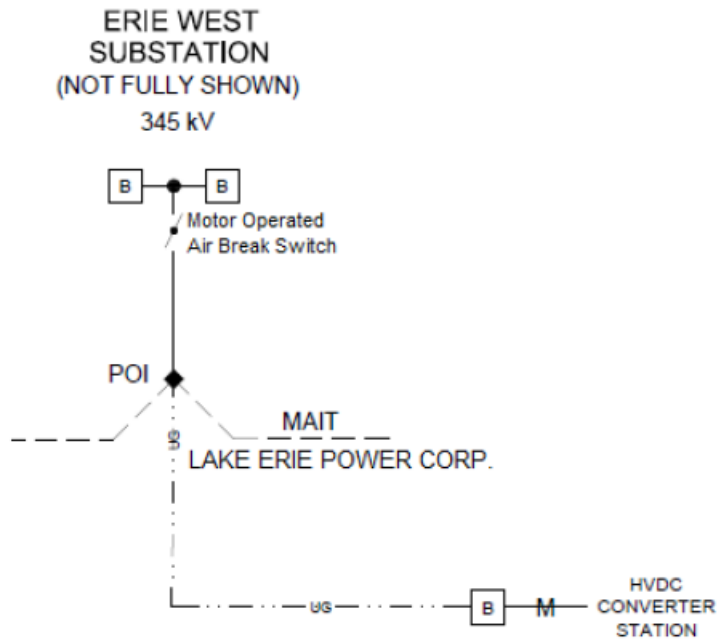
(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

Attachment 1: Project Location



Attachment 2: One Line Diagram



◆ = POINT OF INTERCONNECTION (POI) WILL BE LOCATED WITHIN THE ERIE WEST 345 KV SUBSTATION FENCE. THE POINT BETWEEN THE DEVELOPER-OWNED TERMINATORS ON THE 345 KV UNDERGROUND ATTACHMENT CABLE AND TRANSMISSION OWNER'S TERMINATION CONDUCTORS/STRUCTURE WILL BE DEFINED AS THE POINT OF INTERCONNECTION.

M = REVENUE METERING

Attachment 3: Stability Analysis Results

Executive Summary

PJM queue project AB2-019 is an interconnection request for a Merchant Transmission line to connect Ontario IESO's transmission system to MAIT's transmission system, in Erie County, PA. AB2-019 is an uprate to Y3-092, which was a new interconnection request for adding a HVDC link between Erie West 345 kV substation in the PJM system and Nanticoke 500 kV substation in the IESO system. After AB2-019 uprate, the HVDC link will be based on ± 320 kV, VSC technology for 1032 MW withdrawal from the IESO system with a net 1000 MW injection into the PJM system for the import condition and 1028 MW withdrawal from the PJM system for the export condition from PJM perspective.

Dynamic simulation analysis is required as part of PJM's Generation and Transmission Interconnection Process. This Scope of Work describes how the dynamic simulation analysis of the AB2-019 will be carried out as part of the overall system impact study.

For the import conditions, the HVDC link was set at 1000 MW injection into the PJM system with the corresponding withdrawal from the IESO's system. The AC Control Mode for the VSC on both sides (Nanticoke 500 kV and Erie West 345 kV) was set at voltage.

For the export conditions, the HVDC link was set at 997.7 MW injection into the IESO system with the corresponding withdrawal (1028 MW) from the PJM system. The AC Control Mode for the VSC on both sides (Nanticoke 500 kV and Erie West 345 kV) was set at voltage.

This report describes a dynamic simulation analysis of AB2-019 as part of the overall system impact study.

The load flow scenarios for the analysis were based on the RTEP 2020 Light Load and Summer Peak case, modified to include applicable queue projects.

The AB2-019 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are covered by P1 and P4 contingencies.

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

The three phase faults with normal clearing time will also be performed with prior outage of:

- a) Erie West - Ashtabula - Perry 345kV line
- b) Erie West – Wayne 345kV line
- c) Erie West – Erie South 345kV line
- d) Armstrong – Homer City 345kV line
- e) Wayne – Handsome Shake 345kV line

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the remaining fault contingencies tested on the 2020 Light Load and Summer Peak case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3% for all modes.
- b) The HVDC was able to ride through all faults.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.